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RESEARCH**

APPLICATION NUMBER:

203791Orig1s000

**CLINICAL PHARMACOLOGY AND
BIOPHARMACEUTICS REVIEW(S)**

ONDQA (Biopharmaceutics) Review

Original NDA	203-791 (000)
Applicant:	BioAlliance Pharma
Trademark:	Sitavig (proposed)
Stamp Dates	Original: March 13, 2012; Amendments: June 15, 2012; November 16, 2012
GRMP Date	December 7, 2012
PDUFA Goal date	January 11, 2013
Established Name:	Acyclovir
Dosage Form:	Buccal tablet 50 mg
Route of Administration:	Oral/Buccal
Indication:	Herpes Labialis
Reviewers	Tapash Ghosh, Ph. D. John Duan, Ph. D.

SYNOPSIS

Submission: The Applicant, BioAlliance, submitted a 505(b)(2) NDA application for Sitavig also known as Acyclovir Lauriad™ or ABT 50 mg, a mucoadhesive buccal tablet that provides extended-release of acyclovir in the oral cavity. ABT 50 mg is a white to slightly yellow mucoadhesive buccal tablet with a rounded side and a flat side. Sitavig adheres to the upper gum just above the incisor tooth with the flat surface facing the cheek mucosa for the treatment of recurrent orofacial herpes simplex virus (HSV) infections in immunocompetent patients, (b) (4)

Review: This biopharmaceutics review is focused on the evaluation and approvability of (1) the proposed *in-vitro* dissolution method and acceptance criteria and (2) the Applicant's approach to bridge the products manufactured at (b) (4) and Farnea sites.

In-vitro Dissolution

Dissolution Method: The following proposed dissolution method is acceptable.

Variable	Parameter
Number of Vessels	6
Dissolution Medium	pH 6 phosphate (KH ₂ PO ₄) buffer (0.2M)*
Volume of Dissolution Medium	1000 mL
Temperature of Medium	37.0 ± 0.5°C
USP Apparatus	Baskets
Stirring Speed	60 rpm
Sampling time points**	1.5 and 12 hr

* The pH of the dissolution medium is adjusted to 6.0 ± 0.5

**Extra sampling time points may be performed for example at: 2, 4, 6, 8, 10, 11, 15, 18, 21 and 24 hr for specific studies

The following proposed *in-vitro* dissolution acceptance criteria are also acceptable:

Table 1 Specification Ranges for Dissolution Test	
Time Point (hr)	Ranges (%)
1	(b) (4)
5	
12	

Manufacturing Sites Bridging:

Based on (1) multi-media and multi-point dissolution profiles comparison, and (2) a mathematical analysis of the *in vitro* dissolution profile data and *in vivo* saliva release data from ABT 50 mg manufactured at Farnea and (b) (4) sites are adequately bridged.

RECOMMENDATION

ONDQA-Biopharmaceutics has reviewed the overall dissolution information and from the Biopharmaceutics view point NDA 203-791 for Acyclovir Buccal Tablet 50 mg is recommended for approval.

Tapash K. Ghosh, Ph. D.
Primary Biopharmaceutics Reviewer
Office of New Drug Quality Assessment

Angelica Dorantes, Ph. D.
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BIOPHARMACEUTICS ASSESSMENT

Drug Product:

The next table describes the composition of the formulation for the ABT 50 drug product.

ABT 50 Ingredients

Name of Ingredient	Quantity per Tablet mg (%)
Acyclovir (active ingredient)	50 (b) (4)
Microcrystalline cellulose	(b) (4)
Povidone	
Sodium Lauryl Sulfate	
Hypromellose (b) (4)	
Milk Protein Concentrate (MPC)*	
Magnesium stearate	
Colloidal Silicon Dioxide	
	(b) (4)
Total	(b) (4) (100%)

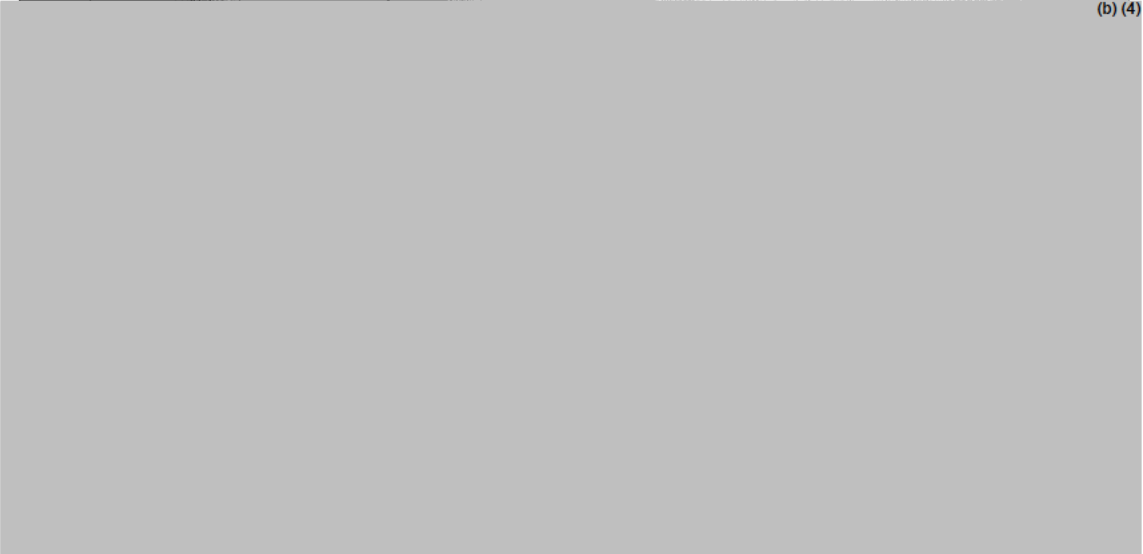
Dissolution Method Development:

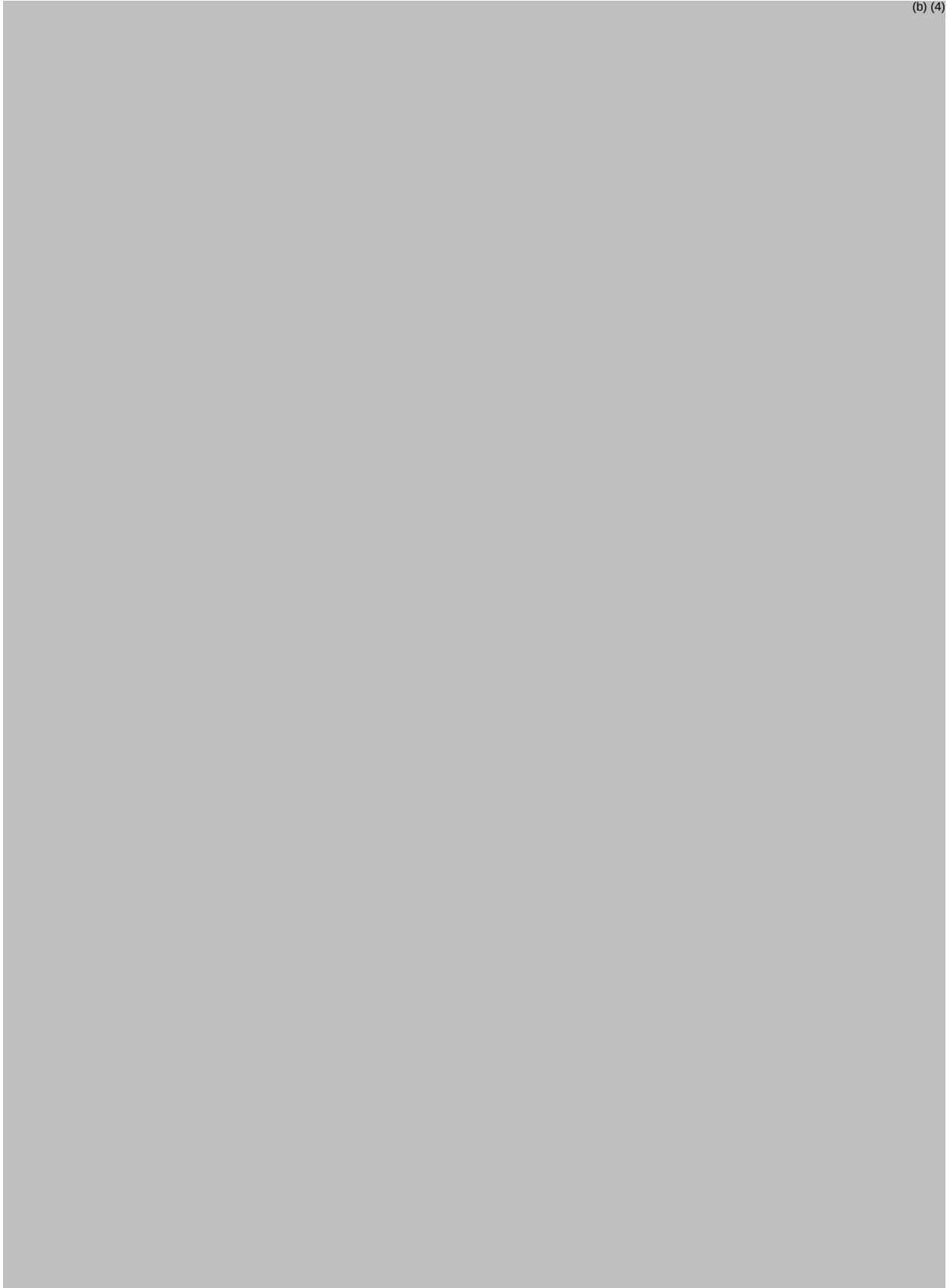
BioAlliance Pharma developed the *in vitro* dissolution method described below with the objective to set *in vitro* drug-release criteria that ensure batch-to- batch and unit-to-unit consistency.

The following parameters were evaluated during the method's development:





Table 8Batches Composition		
		Composition % (w/w)
		



Therefore, the proposed dissolution method can discriminate formulations with variations of some excipients.

Conclusion

Based on the data generated during the development of the method, the proposed dissolution method for acyclovir Lauriad® 50 mg mucoadhesive buccal tablet is as follows:

Variable	Parameter
Number of Vessels	6
Dissolution Medium	pH 6 phosphate (KH ₂ PO ₄) buffer (0.2M)*
Volume of Dissolution Medium	1000 mL
Temperature of Medium	37.0 ± 0.5°C
USP Apparatus	Baskets
Stirring Speed	60 rpm
Sampling time points**	1. 5 and 12 hr

* The pH of the dissolution medium is adjusted to 6.0 ± 0.5

**Extra sampling time points may be performed for example at: 2, 4, 6, 8, 10, 11, 15, 18, 21 and 24 hr for specific studies

HPLC Operating Conditions

Chromatographic description	
Column	(b) (4)
Mobile phase	
Flow rate	
Injection volume	
Wavelength	
Column temperature	
Retention time	
Run time	

Dissolution Acceptance Criteria:

The dissolution test method and acceptance criteria for ABT 50 mg have been developed and validated to assure local control of release of the drug in the oral cavity, over an extended period. ABT 50 mg is a unique locally acting dosage form providing an early and sustained buccal release of acyclovir at the herpes labialis infection site (i.e. the oral mucosa and lip).

The dissolution acceptance criteria originally proposed for ABT 50 mg by the applicant are presented in Table 1 below.

Table 1: Specification Ranges for Dissolution Test	
Time Point (hr)	Ranges (%)
1	(b) (4)
5	
12	

The proposed criteria were based on data points collected on 10 batches as described below. The dissolution data of the product are summarized in Table 2:

- development batches representative of commercial size
- primary stability batches (stability data collected up to the last compliant timepoint at 25°C/60% RH and at 40°C/75% RH).
- supportive stability batches (stability data collected up to the last compliant timepoint at 25°C/60% RH and at 40°C/75% RH).

Table 2. Summary of 10 Batches Dissolution Data

Table 2 Summary of Dissolution Data of ABT 50 mg on 10 available batches			
Dissolution timepoint	1 hr	5 hr	12 hr
Individual Min Value (%dissolved)	5	31	75
Individual Max Value (%dissolved)	20	62	109
Mean (%dissolved)	9	47	96
SD (%)	2	5	6
Mean + 2 S.D.	13	57	108
Mean – 2 S.D.	5	37	84

Three sampling points were recommended for the setting of the dissolution acceptance criteria:

- An early time point (1 hr) is proposed in order to exclude dose dumping. The specification is less than (b) (4) % of the drug dissolved;
- The end point (12 hr) is proposed to ensure that the majority of the active substance has been released (more than (b) (4) % dissolved).

These two points are consistent with the purpose of ABT 50 mg. Indeed, ABT 50 mg has been developed as a local extended-release mucoadhesive buccal tablet in order to optimize local delivery of acyclovir.

- A third point (5 hr) was added in order to ensure compliance with the shape of the dissolution profile (around (b) (4) % dissolved). This time-point was based on the data collected during the development of the product, at release and during stability studies of the drug product. Target dissolution at 5 hr is based on the mean dissolution value of the collected individual data and corresponds to (b) (4) % of dissolved acyclovir.

The range of $\pm$ (b) (4) % applied for this time point is consistent with the individual value repartition around the target and corresponds to $\pm$ (b) (4) standard deviations. This range is also consistent with the local extended release product, meaning high and sustained acyclovir concentrations locally in saliva and lip.

Based on the review of the individual and mean value of the dissolution data from all 10 batches, the reviewer recommended the following dissolution acceptance criteria, which was accepted by the applicant via a submission dated November 16, 2012.

Table 1 Specification Ranges for Dissolution Test		
Time Point (hr)	Ranges (%)	
1		(b) (4)
5		
12		

Bridging Clinical (PK, Pivotal) Batches and Primary Stability and European process Validation Batches

Three batches of ABT 50 mg and 100 mg ((b) (4) and (b) (4) were used for clinical trial purposes. Three batches of ABT 50 mg (DEV7484, DEV7486 and DEV7504) were used for (b) (4) primary stability studies and European process validation. During drug product development, the manufacturing process were transferred from the clinical development site, (b) (4), to the proposed commercial manufacturing site, Farnea. Therefore, three additional batches of ABT 50 mg (two pilot batches 10-81-046 and 10-81-048 and one industrial batch 001) were used for primary stability studies for Farnea site registration. These primary stability batches are representative of the commercial formulation and manufacturing process at the proposed commercial site, Farnea.

The site change is considered a Level 3 change as per SUPAC MR guidance, and that requires either a single-dose bioequivalence (BE) study or an established *in-vitro/in-vivo* correlation (IVIVC) to bridge the formulations manufactured at two different sites. However, in previous communications during pre-NDA stage, it was agreed upon that due to extremely low and variable systemic exposures of acyclovir following the buccal tablet administration, a multi-media and multi-point dissolution comparison coupled with other supportive document like mathematical modeling will be considered in lieu of BE study or IVIVC. As such, the applicant submitted (1) multi-media and multi-point dissolution profiles comparison, and (2) a mathematical analysis of the dissolution profile *in vitro* and saliva release *in vivo*. The mathematical model was developed to demonstrate the relationship of *in vitro* dissolution with *in vivo* saliva dissolution, and to propose a predictive model of the *in vivo* local availability of acyclovir. The purpose of the model is to confirm the sameness of ABT 50 mg manufactured at Farnea with product manufactured at (b) (4).

The two clinical batches containing 50 mg of acyclovir ((b) (4)) and ((b) (4)) the three ((b) (4)) primary stability and European validation batches (DEV7484, DEV7486 and DEV7504) and the three Farnea primary stability batches (10-81-046, 10-81-048 and 001) have been compared in order to show that all primary stability batches and/or European validation batches are representative of the batches used in clinical studies and to support the equivalence between both manufacturing sites. Details of all these batches are summarized in Table 2.

Table 2 : Characteristics of Clinical, (b) (4) Primary Stability and European Validation Batches, and Farnea Primary Stability Batches									
Batch Number	(b) (4)			DEV7484	DEV7486	DEV7504	10-81-046	10-81-048	001
Strength	50 mg	100 mg	50 mg	50 mg			50 mg		
Site of Manufacture	(b) (4)						Farnea		
Batch Size	(b) (4)								
Date of Manufacture	21 February 2005	28 February 2005	17 October 2006	26 May 2010	23 June 2010	28 June 2010	24 August 2010	24 August 2010	04 February 2010
Use	Clinical and Stability	Clinical and Stability	Clinical and Stability	Primary Stability and European Process Validation			Primary Stability		
Clinical Study Number	BA2004/21/01 ^a	BA2004/21/01 ^a	BA2005/21/02 ^b	Not applicable			Not applicable		
^a = Formerly (b) (4)									
^a = PK/PD study in healthy volunteers									
^b = Efficacy study in herpes labialis in immunocompetent patients									

The composition of the clinical batches, primary stability and validation batches are presented in Table 3.

Table 3: Composition of Clinical Batches and Primary Stability and European Validation Batches							
Batch Purpose	Clinical Batches					Primary Stability and European Validation Batches	Primary Stability Batches
Batch Number	(b) (4)					DEV7484	10-81-046
						DEV7486	10-81-048
						DEV7504	001
Manufacturer		(b) (4)					Farmea
Component	% (w/w)	mg/tablet	mg/tablet	mg/tablet	% (w/w)	mg/tablet	mg/tablet
Acyclovir	(b) (4)	50.000	100.000	50.000	(b) (4)	50.000	50.000
Povidone	(b) (4)	(b) (4)					
Microcrystalline cellulose							
Sodium Lauryl Sulfate							
Hypromellose type (b) (4)							
MPC							
Magnesium stearate							
Colloidal silicon dioxide							
Total	100.00	(b) (4)			100.00	(b) (4)	
(b) (4)							

Comparison of Dissolution Profiles in Multi-media:

The similarity of the dissolution profiles was shown between both formulation (b) (4) versus (b) (4) 0% of magnesium stearate) when tested at different pH (4, 6 and 7).

The dissolution profiles at pH6 of clinical batches (b) (4) and (b) (4) and primary stability and validation batches (DEV7484, DEV7486 and DEV7504) of ABT 50 mg are presented in Figure 1 and Figure 2.

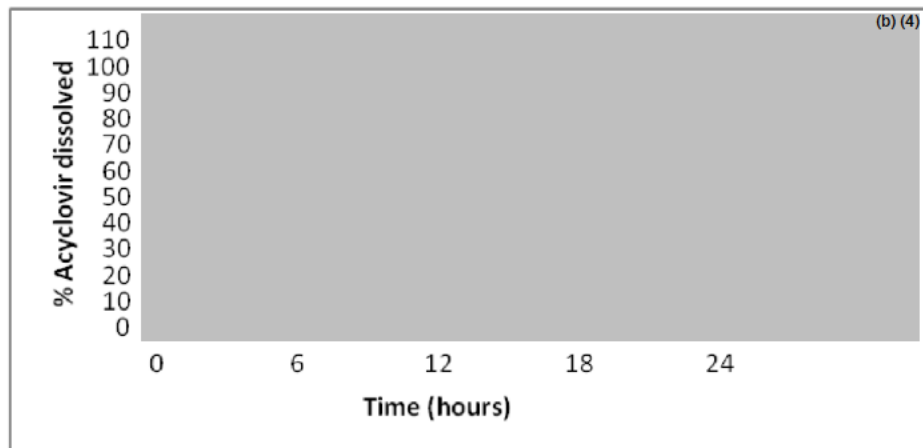


Figure 1 Dissolution profiles of tablets manufactured at the clinical development site and at the commercial production site.

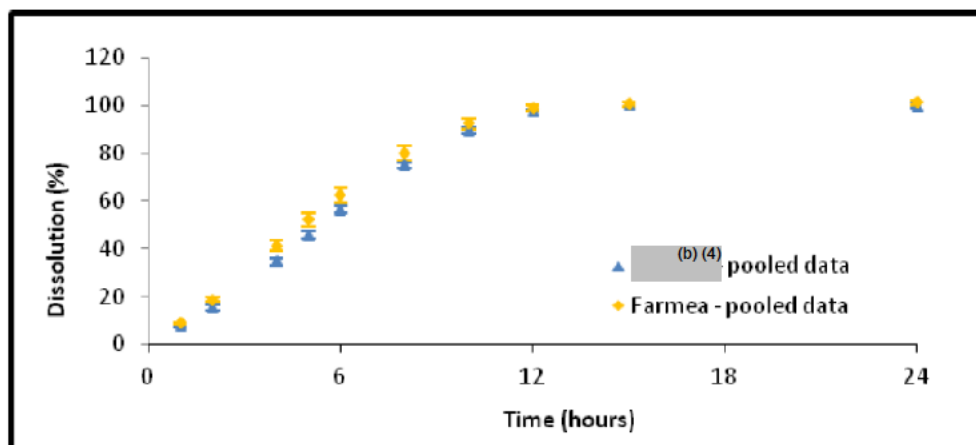


Figure 2 Dissolution profiles of tablets manufactured at Farmea and (b) (4) (pooled data)

The results indicate that the dissolution profiles are the same as indicated by the f_2 , similarity factors of greater than 50.

In order to further demonstrate the similarity of ABT 50 mg manufactured at (b) (4) and Farmea, ABT 50 mg dissolution profiles have been studied in different media covering the reported pH range of saliva.

Multipoint dissolution profiles were generated in four different bio-relevant media at three different pH values (4, 6 and 7) and in simulated salivary fluid (pH 6.8). Dissolution testing was performed on 12 tablets over 15 hours for similarity purposes.

The dissolution profiles in pH 4, 6 and 7, and in simulated saliva fluid of batches manufactured at both Farmea and (b) (4) sites are presented in Figure 3, Figure 4, Figure 5 and Figure 6.

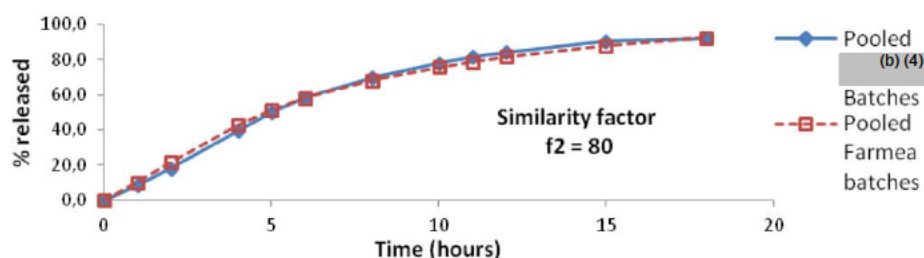


Figure 3: Dissolution profiles of pooled (b) (4) Batches (Primary stability and European validation batches DEV7504, DEV7484, DEV7486) versus pooled Farmea batches (Primary stability batches 10.81.046, 10.81.048, 001) at pH 4

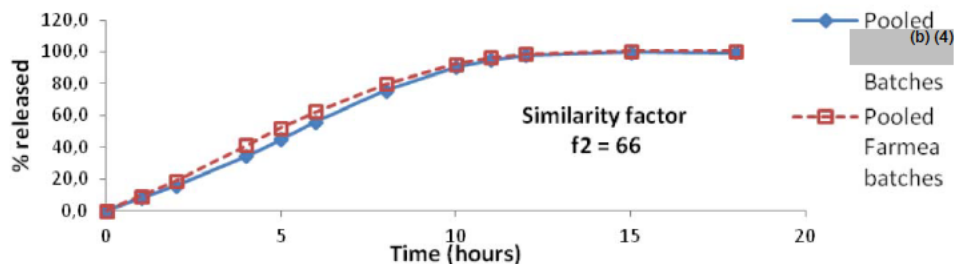


Figure 4: Dissolution profiles of pooled (b) (4) Batches (Primary stability and European validation batches DEV7504, DEV7484, DEV7486) versus pooled Farmea batches (Primary stability batches 10.81.046, 10.81.048, 001) at pH 6

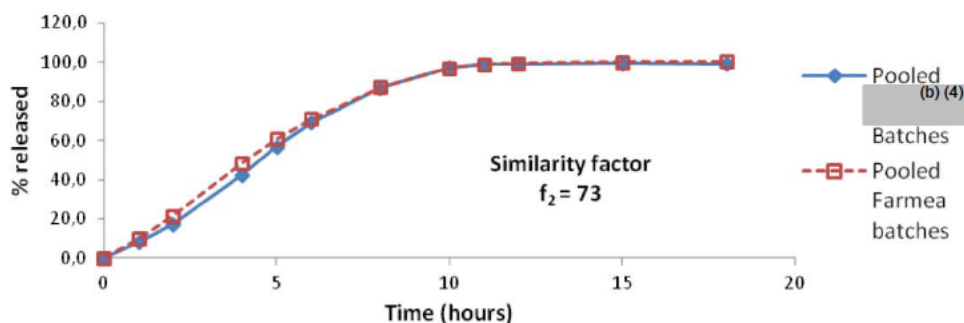


Figure 5: Dissolution profiles of pooled (b) (4) Batches (Primary stability and European validation batches DEV7504, DEV7484, DEV7486) versus pooled Farmea batches (Primary stability batches 10.81.046, 10.81.048, 001) at pH 7

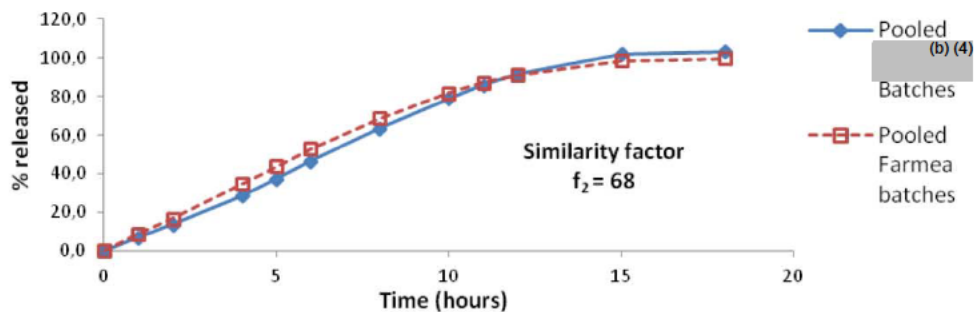


Figure 6: Dissolution profiles of pooled (b) (4) Batches (Primary stability and European validation batches DEV7504, DEV7484, DEV7486) versus pooled Farmea batches (Primary stability batches 10.81.046, 10.81.048, 001) in simulated salivary fluid

The similarity factors, f_2 , were calculated for manufacturing site comparison using the dissolution data collected in the multi-media dissolution experiments. The results are provided in Table 4 below.

Table 4: Similarity factor calculated for multi-media dissolution between (b) (4) (pooled primary stability and European validation batches) and Farmea (pooled primary stability batches)				
Dissolution media	pH 4	pH 6	pH 7	Simulated saliva fluid
f_2	80	66	73	68

In conclusion, similarity between the batches manufactured at (b) (4) and the batches produced at Farnea was obtained in all bio-relevant dissolution media, further supporting the sameness of ABT 50 mg produced at each of the two manufacturing sites.

Mathematical Modeling:

A mathematical analysis of the dissolution profile *in vitro* and *in vivo* was performed. The mathematical model was developed to demonstrate the relationship of *in vitro* dissolution with *in vivo* saliva dissolution, and to propose a predictive model of the *in vivo* local availability of acyclovir. The purpose of the model was to confirm the sameness of ABT 50 mg manufactured at Farnea with product manufactured at (b) (4)

The review by Dr. John Duan of the mathematical analysis is presented in pages 16 to 22 of this review.

ONDQA BIOPHARMACEUTICS REVIEW

NDA#:	203-791/S-000
Submission Date:	3/12/2012
Drug Name:	Sitavig (Acyclovir) Buccal Tablets
Formulation:	Buccal Tablets
Strength:	50 mg
Applicant:	BioAlliance Pharma
Reviewer:	John Duan, Ph.D.
Submission Type:	Site Change Issue

SYNOPSIS

Based on the proposed indication and formulation characteristics, the use of an *in vitro* release profile comparison method to show the formulation similarity between the two sites is considered appropriate.

COMMENTS

1. The proposed buccal formulation shows higher and more sustained concentrations in the saliva compared to its systemic concentration, which supports the claimed release characteristics of the formulation and the rationale of measuring the saliva concentration. Therefore, correlating the saliva concentration with *in vitro* dissolution is a reasonable approach.
2. The *in vitro* dissolution profiles are parallel to the *in vivo* saliva release. The *in vitro* dissolution profiles are considered to be a surrogate for the *in vivo* saliva release profiles.
3. The similarity between the batches manufactured at (b) (4) and Farnea sites (b) (4) can be established based on the *in vitro* dissolution profile comparisons.

RECOMMENDATION

It is recommended that the *in vitro* dissolution profile comparisons be used to show the similarity between the batches manufactured at (b) (4) and Farnea sites.

APPENDIX. EVALUATIONS OF THE MODELING APPROACH

I. Background

The proposed Acyclovir Lauriad™ 50 mg mucoadhesive Buccal Tablets (ABT 50 mg) have been developed for the local treatment of orofacial herpes (b) (4)

During drug product development, the manufacturing process was transferred from the clinical development site, (b) (4) (b) (4), to the proposed commercial manufacturing site, Farnea (France).

Three batches of ABT 50 mg (DEV7484, DEV7486 and DEV7504) were used for (b) (4) primary stability studies and European process validation. Three additional batches of ABT 50 mg (two pilot batches 10-81-046 and 10-81-048 and one industrial batch 001) were used for primary stability studies for Farnea site registration. These primary stability batches are representative of the commercial formulation and manufacturing process at the proposed commercial site, Farnea.

The two clinical batches containing 50 mg of acyclovir (b) (4) and (b) (4) the three (b) (4) primary stability and European validation batches (DEV7484, DEV7486 and DEV7504) and the three Farnea primary stability batches (10-81-046, 10-81-048 and 001) have been compared in order to show that all primary stability batches and/or European validation batches are representative of the batches used in clinical studies and to support the equivalence between both manufacturing sites.

Details of all these (b) (4) batches are summarized in the following table.

Characteristics of Clinical, (b) (4) Primary Stability and European Validation Batches, and Farnea Primary Stability Batches									
Batch Number	(b) (4)			DEV74 84	DEV74 86	DEV75 04	10-81-046	10-81-048	001
Strength	50 mg	100 mg	50 mg	50 mg			50 mg		
Manufacture Site	(b) (4)						Farnea		
Batch Size	(b) (4)								
Date of Manufacture	2 February 2005	28 February 2005	17 October 2006	26 May 2010	23 June 2010	28 June 2010	24 August 2010	24 August 2010	4 February 2010
Use	Clinical and Stability	Clinical and Stability	Clinical and Stability	Primary Stability and European Process Validation			Primary Stability		
Clinical Study Number	BA2004/21/01 ^a	BA2004/21/01 ^a	BA2005/21/02 ^b	Not applicable			Not applicable		
* = Formerly (b) (4)									
a = PK/PD study in healthy volunteers									
b =Efficacy study in herpes labialis in immunocompetent patients									

The Applicant stated that ABT 50 mg is a local drug delivery product that results in a delayed and very low systemic exposure; therefore, no *in vitro in vivo* correlation can be established. A mathematical analysis of the dissolution profile *in vitro* dissolution and *in vivo* saliva release was performed. The mathematical model was developed to demonstrate the relationship of *in vitro* dissolution with *in vivo* saliva release, and to propose a predictive model of the *in vivo* local availability of acyclovir. The purpose of

the model is to confirm the sameness of ABT 50 mg manufactured at Farmea with product manufactured at (b) (4)

The *in vitro* dissolution profile has been analyzed and exhibits triphasic release where a first very small fraction of the dose exhibits a sigmoid process simultaneously with a principle zero order slow release process (major part of released dose), and a terminal third phase corresponding to a delayed first order phase.

The development of the proposed model has been conducted in five main steps:

- Mathematical analysis of the *in vitro* dissolution profile for the clinical batch (b) (4). The dissolution profiles were obtained using the following condition.

Variable	Parameter
Number of Vessels	6
Dissolution Medium	pH 6 phosphate buffer
Volume of Dissolution Medium	1,000 mL
Temperature of Medium	37.0 ± 0.5°C
USP Apparatus	Baskets
Stirring Speed	60 rpm

- Mathematical analysis of the *in vivo* buccal release profile for the clinical batch (b) (4) in a clinical study BA2004/21/01.
- Comparison of the *in vitro* dissolution and *in vivo* release profiles for the clinical batch (b) (4) – introduction of a Lag Time;
- Determination of predicted *in vivo* saliva concentration according to the model for the clinical batch (b) (4) – calculation of predictability error;
- Application of the model for the others batches: the clinical batch (b) (4) and the US primary stability batches (10-81-046; 10-81-048; 001).

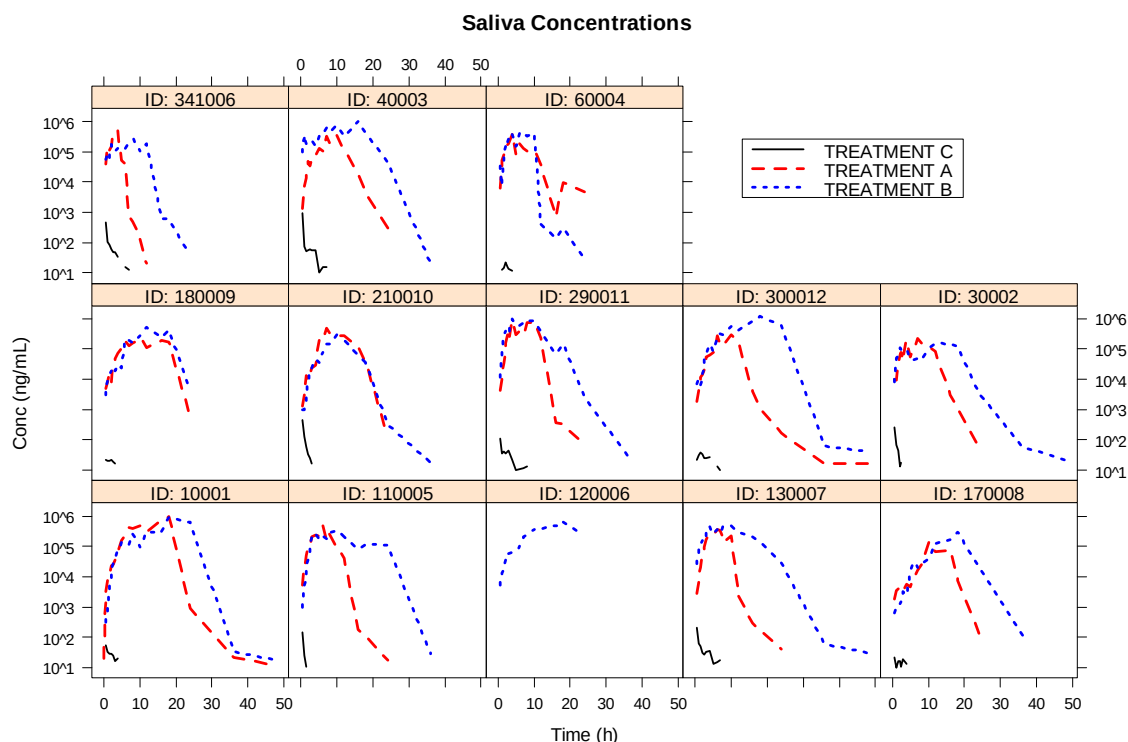
The focus of this section of the review is the evaluation and acceptability of the proposed mathematical model used to justify the manufacturing site change for Acyclovir buccal tablets.

II. Evaluation

1. Comparison between the concentrations in the saliva and in the plasma

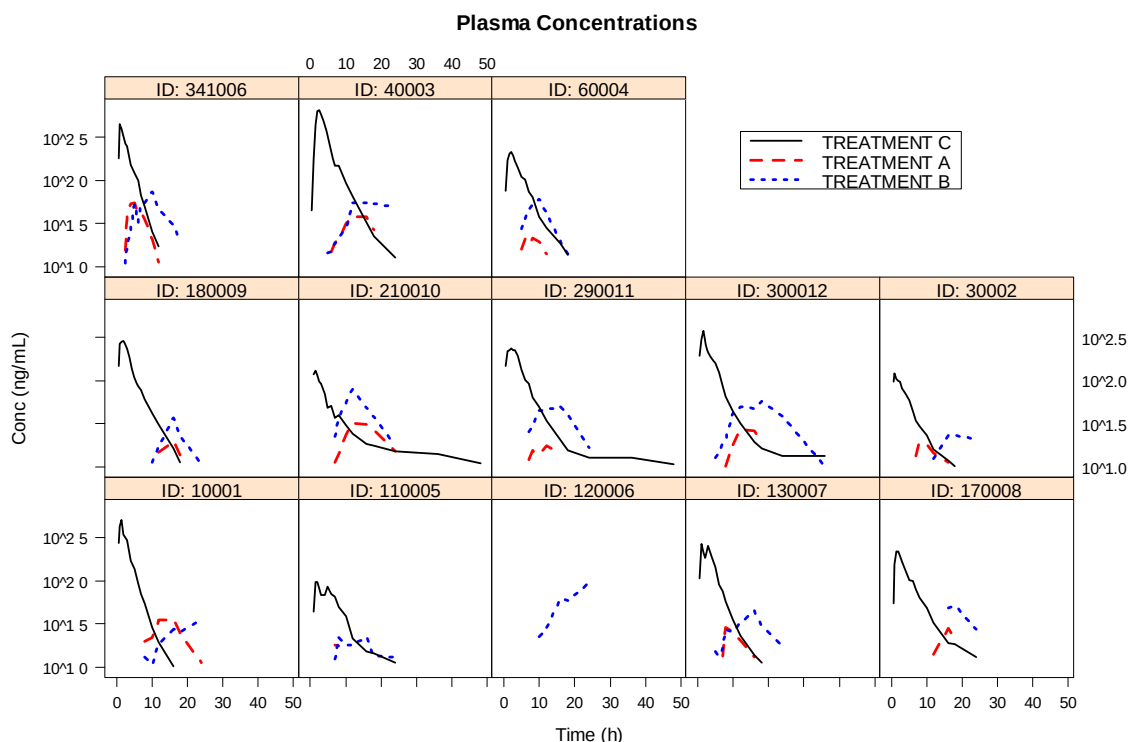
The evaluation began with the identification of the formulation characteristics by comparing between the concentrations in the saliva and in the plasma for the proposed formulation.

The following figure shows the saliva concentration for the 12 subjects (clinical study BA2004/21/01) with three different treatments (an extra subject with only one treatment). Treatment A is 50 mg dose of acyclovir formulated as acyclovir Lauriad® (50 mg) MBT, Treatment B is 100 mg dose of acyclovir formulated as acyclovir Lauriad® (100 mg) MBT, and Treatment C is 200 mg dose of acyclovir formulated as a 200 mg oral tablet (Zovirax®).



As shown, the proposed formulation had much higher concentrations in the saliva compared to the oral formulation, even when the dose of oral formulation is higher (200 mg vs. 50 and 100 mg, respectively).

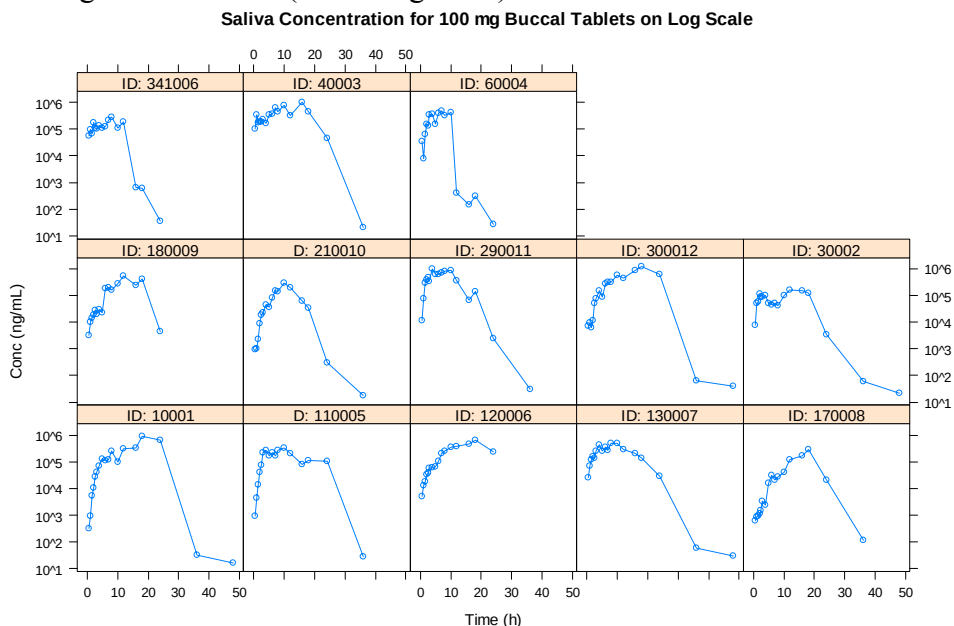
On the other hand, plasma concentrations were lower for the proposed formulation comparing to the oral formulation as shown in the following figure.



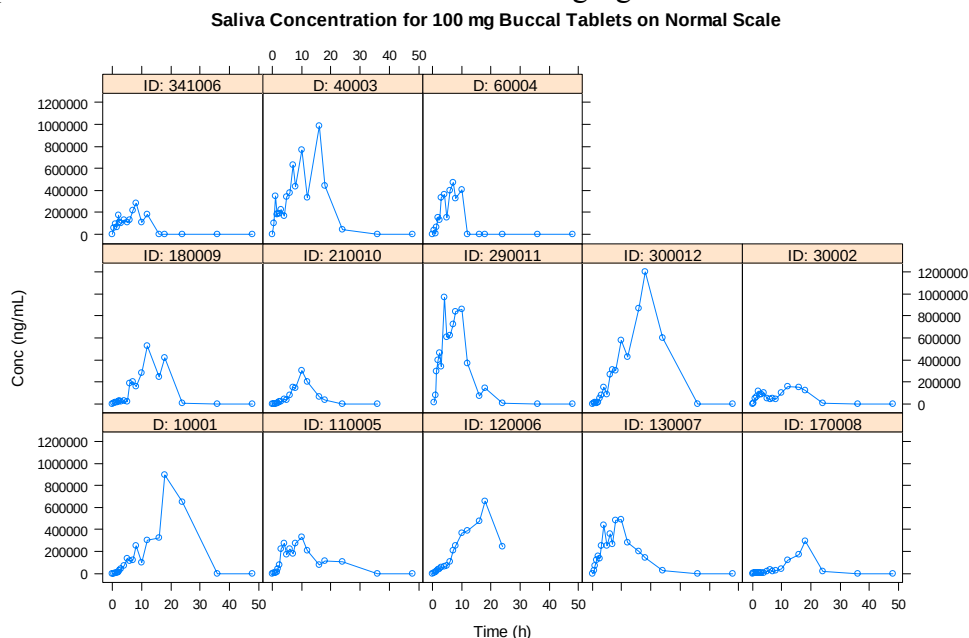
This comparison indicates that the proposed buccal formulation has higher and more sustained concentrations in the saliva compared to its systemic exposure, which supports the claimed release characteristics of the proposed formulation and the rationale of measuring the saliva concentration. Therefore, correlating the saliva concentration with in vitro dissolution is a reasonable approach.

2. The duration of the release in saliva

As shown in the following figure, the most saliva concentrations for the 12 subjects after taking 100 mg buccal tablets (on the log scale) were collected until 24 hours.



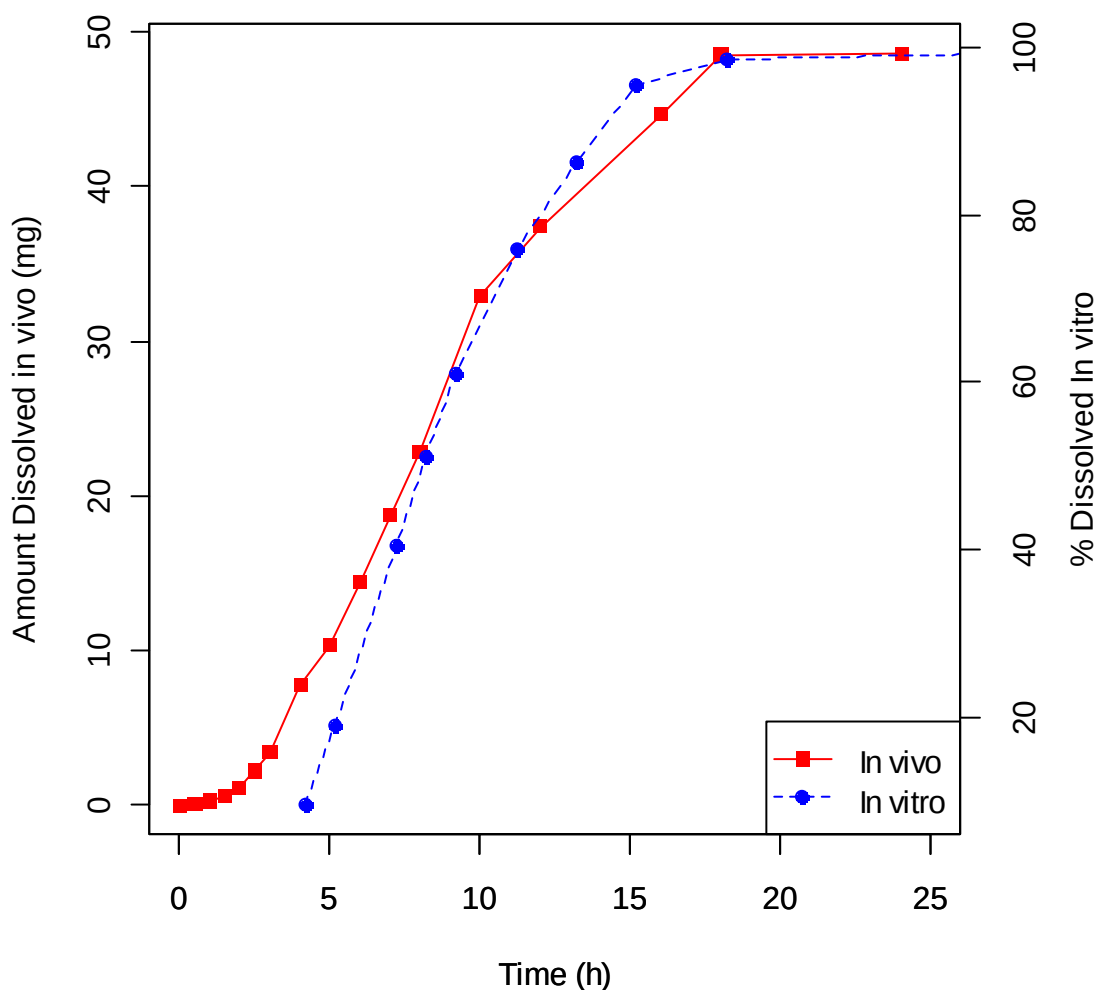
To make sure this sampling time is long enough for the drug release in saliva, the scale of these plots were modified as shown in the following figure that is on the normal scale.



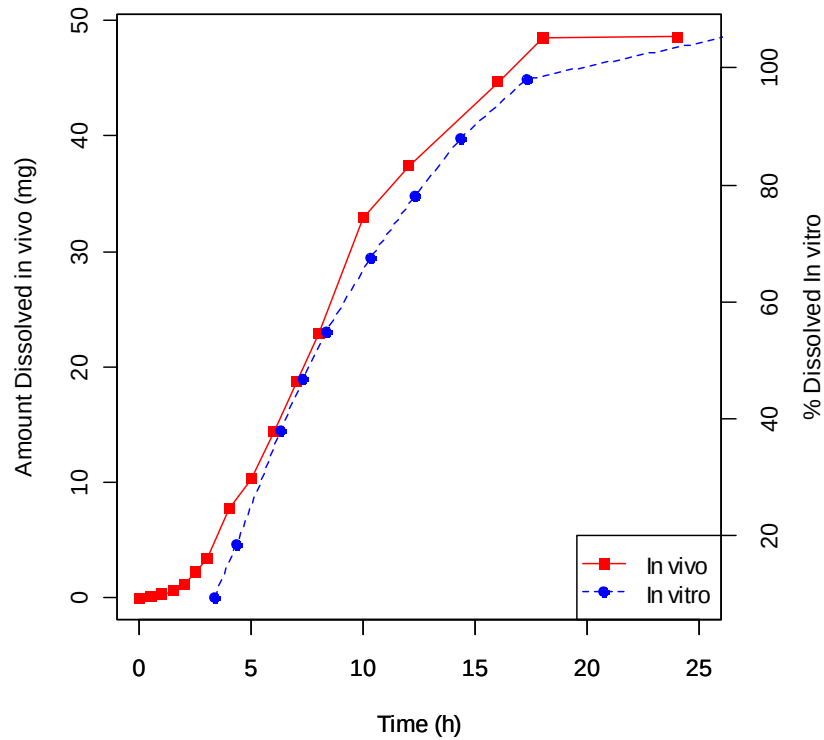
It can be seen, from these profiles, that on average it is reasonable to assume a complete saliva release within 24 hours, which is one of the important assumption for the modeling exercises.

3. The relationship between *in vitro* dissolution and *in vivo* saliva release

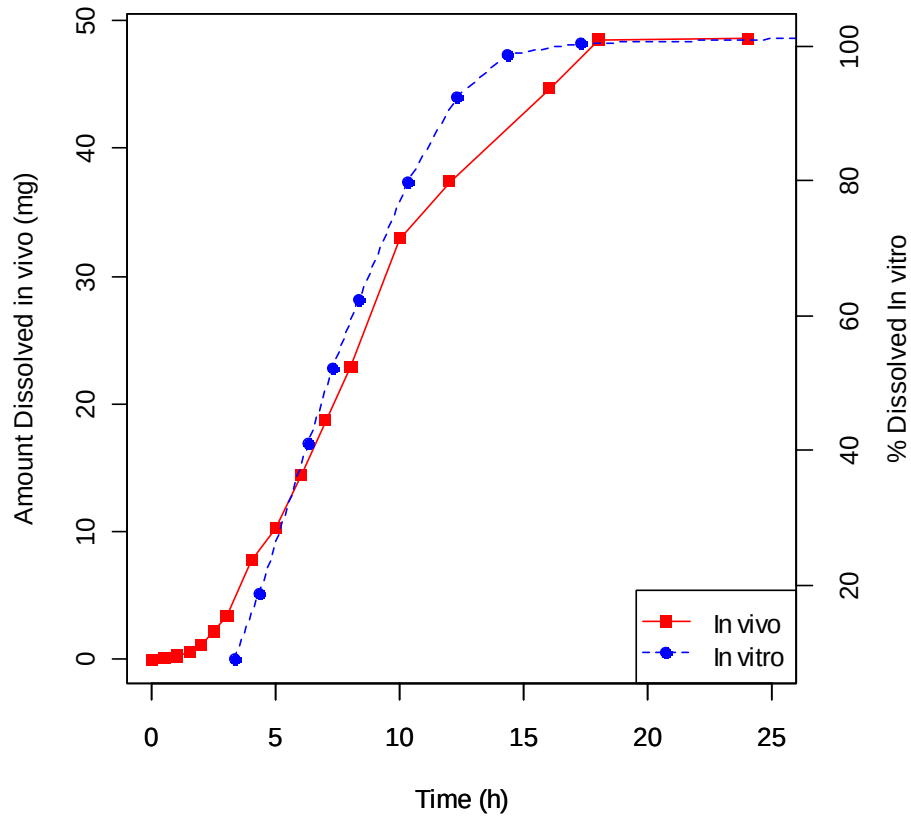
After confirming the release characteristics of the proposed buccal formulation and the assumption that the saliva release is completed within 24 hours, we begin to explore the relationship between the *in vitro* dissolution profile and the *in vivo* saliva release profiles. The approach we are taking is to simply compare the profiles between these two occasions. The following figure shows the overlap between the *in vitro* dissolution (expressed as % dissolved) for lot (b) (4) and the accumulated amount released in saliva obtained from clinical study BA2004/21/01.



The following figure shows a similar plot for the *in vitro* release profile of lot (b) (4) and the accumulated amount released in saliva obtained from clinical study BA2004/21/01.



The following figure shows a similar plot for the mean of the in vitro release for the stability batches (batch # 10-81-046, 10-81-048 and 001, manufactured at Farnea site) and the accumulated amount released in saliva obtained from clinical study BA2004/21/01.



All these plots show a similarity between the *in vitro* dissolution and the *in vivo* saliva release judged by the f2 factor reported below.

Batch #	Reference	f2
% Dissolved <i>in vitro</i> (b) (4) (manufactured at (u) (4) site)	% Released in Saliva in Study BA2004/21/01	79.55
% Dissolved <i>in vitro</i> (b) (4) (manufactured at (u) (4) site)	% Released in Saliva in Study BA2004/21/01	85.11
% Dissolved <i>in vitro</i> mean of the stability batches (# 10-81-046, 10-81-048 and 001 manufactured at Farnea site)	% Released in Saliva in Study BA2004/21/01	54.32

Also, they indicate that the *in vitro* dissolution is parallel to the *in vivo* release. Therefore, there is enough confidence to conclude that the *in vitro* dissolution reflects the *in vivo* saliva release profiles and the *in vitro* dissolution profile is considered to be a surrogate for the *in vivo* saliva release profile.

In conclusion, this biopharmaceutical modeling approach demonstrates that *in vitro* and *in vivo* release of acyclovir is governed by the same kinetic parameters, both exhibiting a major linear release phase (zero order). The simulation with the mathematical model has shown that predictive *in vivo* local exposure is equivalent for Farnea and (b) (4) batches thus supporting the sameness of ABT 50 mg manufactured at either site.

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/s/

TAPASH K GHOSH
12/11/2012

ANGELICA DORANTES
12/11/2012

OFFICE OF CLINICAL PHARMACOLOGY (OCP) REVIEW

NDA	203791
Submission Date	12 Mar 2012
Brand Name	Sitavig Lauriad®
Generic Name	Acyclovir
Primary Reviewer	Leslie Chinn, Ph.D.
Secondary Reviewer	Shirley Seo, Ph.D.
OCP Division	DCP4
OND Division	DAVP
Applicant	BioAlliance Pharma
Formulation; Strength	Mucoadhesive buccal tablet; 50 mg
Proposed Dosing Regimen	50 mg single dose
Proposed Indication	Local treatment of orofacial herpes and (b) (4)
Review Type	505(b)(2) New Drug Application, standard review

TABLE OF CONTENTS

1. EXECUTIVE SUMMARY	2
1.1. INTRODUCTION	2
1.2. POSTMARKETING COMMITMENTS OR REQUIREMENTS	2
1.3. CLINICAL PHARMACOLOGY SUMMARY	2
1.4. CLINICAL PHARMACOLOGY RECOMMENDATION	3
2. TRIAL BA2004/21/01	3
2.1. DESIGN	3
2.2. RESULTS	4
2.3. DISCUSSION	9
3. TRIAL BA2005/21/02 (Pivotal Efficacy Trial Supporting this Application)	10
3.1. DESIGN	10
3.2. RESULTS	11
3.3. DISCUSSION	13
4. LABELING RECOMMENDATIONS	14
5. CONCLUSIONS	14

1. EXECUTIVE SUMMARY

1.1. INTRODUCTION

This memorandum documents the clinical pharmacology review for NDA 203791, a 505(b)(2) application for a mucoadhesive buccal tablet containing the antiviral drug acyclovir (acyclovir Lauriad™). The indication proposed by the Applicant is for the local treatment of orofacial herpes (i.e. coldsores) and the (b) (4).

Acyclovir is currently marketed as tablets, topical cream, intravenous injection, and ophthalmic ointment for the treatment of herpes simplex, varicella zoster, and herpes zoster virus infections. The topical cream is primarily used for the local treatment of orofacial herpes labialis; there is a topical cream preparation containing acyclovir and hydrocortisone that is approved for the early treatment of the signs and symptoms of recurrent herpes labialis (Xerese®). The Lauriad™ delivery system is a buccal tablet intended to facilitate rapid and prolonged release in the buccal cavity. A Lauriad™ formulation of miconazole (Oravig®) was approved for the treatment of oropharyngeal candidiasis in the US in 2010.

The Applicant submitted data from two studies, a Phase 3 pivotal study (BA2005/21/02) and a pharmacokinetic study (BA2004/21/01), in support of the 505(b)(2) application. In pre-NDA discussions with the Applicant, the Division specified that the primary endpoint should be the time to healing (TTH) of the primary vesicular lesion and that TTH following administration of acyclovir Lauriad™ should be statistically significant and clinically meaningful (at least 0.5 day benefit) compared to placebo. Following evaluation of the results from BA2005/21/02, the clinical and statistical reviewers of this application determined that the prespecified minimum criteria for approval of acyclovir Lauriad™ was not met by the contents of this application; therefore, the Division is not recommending approval of acyclovir Lauriad™ based upon the data contained within the current 505(b)(2) submission. However, from a clinical pharmacology perspective, the information submitted would not preclude this product from being approved.

1.2. POSTMARKETING COMMITMENTS OR REQUIREMENTS

Not applicable.

1.3. CLINICAL PHARMACOLOGY SUMMARY

In addition to dissolution data, the Applicant submitted the results from the pharmacokinetic study BA2004/21/01 in support of a request for a biowaiver for the approval of a change in commercial manufacturing site. The request for a biowaiver was based on the infeasibility of demonstrating bioequivalence in a clinical trial due to low circulating levels of acyclovir in plasma as well as the large intersubject variability in saliva concentrations. A meeting between the Applicant and the Division regarding the biowaiver was held on 26 May 2011 and the request for the biowaiver was deemed appropriate by the Division in a communication sent to the Applicant on 1 Dec 2011, although a final determination of the acceptability of a biowaiver was considered to be a review issue under the NDA.

The Phase 3 pivotal study BA2005/21/02 included the collection of saliva samples for acyclovir quantification as a secondary endpoint. However, in the clinical study report, the Applicant notes that the planned correlation between viral load or efficacy and saliva acyclovir concentrations could not be performed due to limited data. These saliva concentration data were therefore not evaluated in the clinical pharmacology review of this application. In the absence of concentration data, tablet adhesion time in the buccal cavity (coupled with dissolution data) was used to estimate the total amount of acyclovir delivered to the site of action and is described in this review.

1.4. CLINICAL PHARMACOLOGY RECOMMENDATION

The Office of Clinical Pharmacology has reviewed the information submitted to NDA 203-791. The application is approvable from a clinical pharmacology perspective.

2. TRIAL BA2004/21/01

Pharmacokinetic and Tolerability Study of Single Administration of Acyclovir Lauriad® (50 and 100 mg), Mucoadhesive Buccal Tablet (MBT), Compared with Single Administration of Oral Tablet of Acyclovir (200 mg) in Healthy Volunteers

Study Center: aster s.a.s. (Paris, France)

Study Dates: 29 Mar – 27 Jun 2005

Report Date: 7 Oct 2010

2.1. DESIGN

This trial was an open-label, randomized, three-way crossover, single-dose trial in 12 healthy males and females between the ages of 18 and 40 years. There were three treatment periods of two days each, separated by washout periods of at least 7 days.

Treatment A – 50 mg acyclovir (Lauriad® MBT; (b) (4) Developpement Lot (b) (4) 6014)

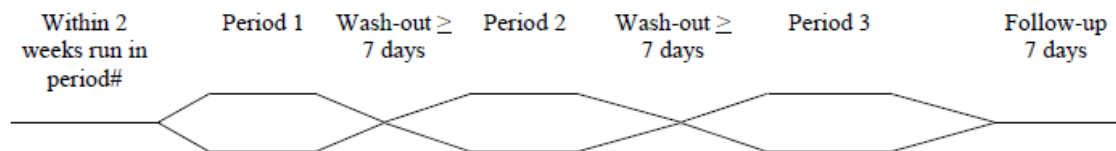
Treatment B – 100 mg acyclovir (Lauriad® MBT; (b) (4) Developpement Lot (b) (4) 6015)

Treatment C – 200 mg acyclovir (Zovirax® tablet; GlaxoSmithKline Lot A-90)

Treatments were administered in the morning following a fast of at least 10 h. Treatment C was administered with 240 mL of water. No additional fluid intake was allowed until after the 4 h blood draw. Standard lunch and dinner were served after the 4 and 10 h blood samples, respectively. Consumption of beverages with alcohol or caffeine was not permitted during the study.

The study schema is shown below.

Figure 1: Study schema (source: Study Report Figure 1)



#: There was a minimum of one-week delay between screening and first dosing

Source: Appendix 16.1.1.1.

All subjects underwent serial blood and saliva sampling for acyclovir plasma and saliva concentrations predose and 0.5, 1, 2, 2.5, 3, 4, 5, 6, 7, 8, 10, 12, 16, 18, 24, 36, and 48 h postdose. In addition, half of the subjects underwent labial stripping predose and 3, 12, and 24 h postdose (Group 1) and half of subjects underwent labial stripping predose and 6, 18, and 24 h postdose (Group 2).

The reported mean terminal half-life of acyclovir in plasma is 2-3 h; the collection of samples to 48 h postdose was therefore likely to be sufficient to characterize the full concentration-time profile of acyclovir in plasma. The mean duration of adhesion of miconazole Lauriad® is 15 h in human clinical trials.

The use of any prescription or over-the-counter drug, with the exception of paracetamol, within one week prior to the first dose of study drug was not permitted. In addition, the use of any drug known to affect hepatic metabolism within one month prior to the first dose of study drug, any drug known to affect renal tubular secretion within two weeks prior to the first dose of study drug, any drug known to affect gastric acid production within two weeks prior to the first dose of study drug, or any drug known to affect gastrointestinal motility within two weeks prior to the first dose of study drug was not permitted.

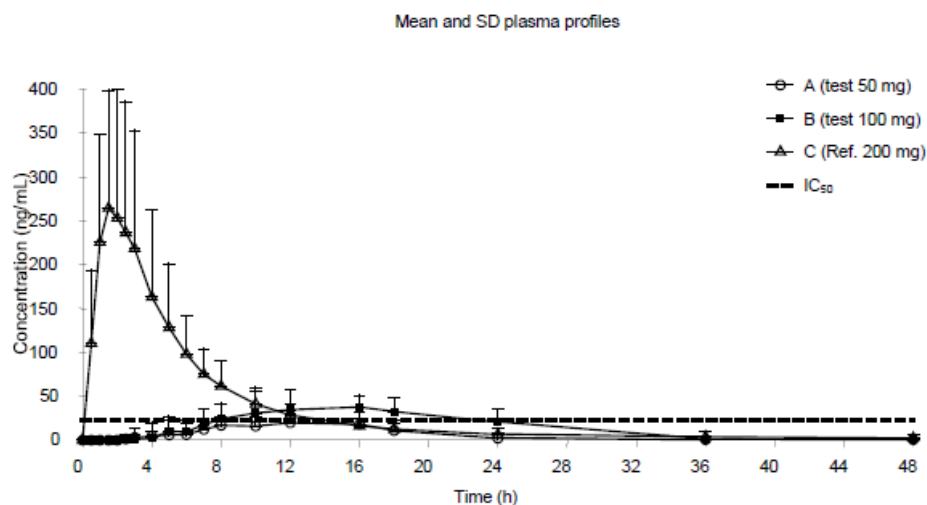
2.2. RESULTS

Thirteen subjects were randomized and received study drug; one subject withdrew for personal reasons on Day 2 of the study, leaving 12 subjects in the pharmacokinetic analysis set. Overall, 76.9% of subjects in the safety analysis set were male and 53.8% of subjects were black, with the remainder Caucasian. The mean age was 27.4 years, with a range of 21 to 37. Two subjects were smokers (smoking a mean of 2 and 4 cigarettes per day).

Acyclovir concentrations in plasma, saliva, and labial strips were determined by a validated high-performance liquid chromatographic (HPLC) method. Pharmacokinetic parameters were calculated using noncompartmental methods (WinNonlin, Pharsight Corporation). The lower limit of quantification was 10 ng/mL for plasma and saliva and 10 ng/2 strips for labial stripping. C_{max} , t_{max} , $t_{1/2}$, AUC_{τ} , AUC_{inf} , and relative bioavailability were determined from plasma concentrations. C_{max} , t_{max} , and AUC_{τ} were determined from saliva concentrations. Tissue concentrations (to be compared to the acyclovir IC_{50}) were determined from labial strips.

The plasma concentration-time profiles of the three acyclovir treatments are shown in Figure 2. Compared to the oral tablet, the Lauriad® formulations resulted in delayed absorption (median t_{lag} of 6 h) and markedly lower plasma concentrations (geometric mean values of C_{max} were 90 and 80% lower and geometric mean values of AUC_{tau} were 88% and 65% lower for the 50 and 100 mg MBT, respectively; see Table 1). Estimations of AUC_{inf} and $t_{1/2}$ were not possible in the majority of subjects due to low and erratic acyclovir concentrations at later timepoints. The relative bioavailabilities (based on AUC_{tau}) of acyclovir Lauriad® 50 and 100 mg tablets compared to the acyclovir oral tablet were 0.12 and 0.35, respectively.

Figure 2. Mean $\pm$ SD plasma concentration-time profiles after a single dose of acyclovir 200 mg oral tablet or Lauriad® 50 or 100 mg MBT (source: Study Report Figure 3)



Source: Table 14.4.1 / Appendices 16.2.10.1

Table 1. Acyclovir plasma pharmacokinetic parameters after a single dose of acyclovir 200 mg oral tablet or Lauriad® 50 or 100 mg MBT (source: Study Report Table 6)

Treatment		t_{lag} (h)	t_{max} (h)	C_{max} (ng/mL)	AUC_{0-t} (h*ng/mL)	$AUC_{0-\infty}$ (h*ng/mL)	$t_{1/2}$ (h)	Frel*
Acyclovir Lauriad® A: 50 mg	N	12	12	12	12	4	4	12
	Mean	6.25	11.33	28.48	225	390	6.32	0.15
	SD	2.22	4.05	10.95	132	138	2.73	0.09
	Min	2	5	17.43	27	290	2.6	0.03
	Median	6	12	27.69	170	341	6.88	0.11
	Max	10	16	55.29	422	590	8.92	0.35
	Geo. Mean	5.82	10.61	26.83	182	375	5.74	0.12
	CV%	36	36	38	59	35	43	61
Acyclovir Lauriad® B: 100 mg	N	12	12	12	12	4	4	12
	Mean	5.83	15.5	49.22	567	1758	17.31	0.37
	SD	2.59	4.01	17.88	250	1957	24.63	0.13
	Min	2	10	22.35	288	581	3.8	0.2
	Median	6	16	51.17	524	883	5.61	0.36
	Max	12	24	78.62	1096	4685	54.23	0.66
	Geo. Mean	5.33	15.02	45.92	520	1206	8.96	0.35
	CV%	44	26	36	44	111	142	36
Ref. Zovirax® C: 200 mg	N	12	12	12	12	12	12	
	Mean	0.08	1.63	295.95	1603	1997	23.43	
	SD	0.19	0.48	168.44	682	918	31.61	
	Min	0	1	97.71	711	851	2.53	
	Median	0	1.5	250.33	1499	1663	7.88	
	Max	0.5	2.5	647.11	3359	3518	101.66	
	Geo. Mean	NC	1.56	254.09	1486	1822	11.5	
	CV%	234	30	57	43	46	135	
A vs C	PE** (90% CI)		p<0.05#	0.11 (0.08-0.13)	0.12 (0.09-0.16)			
B vs C	PE (90% CI)		p<0.05#	0.18 (0.14-0.23)	0.35 (0.26-0.47)			
A vs B	PE (90% CI)		p<0.05#	0.59 (0.46-0.74)	0.35 (0.26-0.47)			

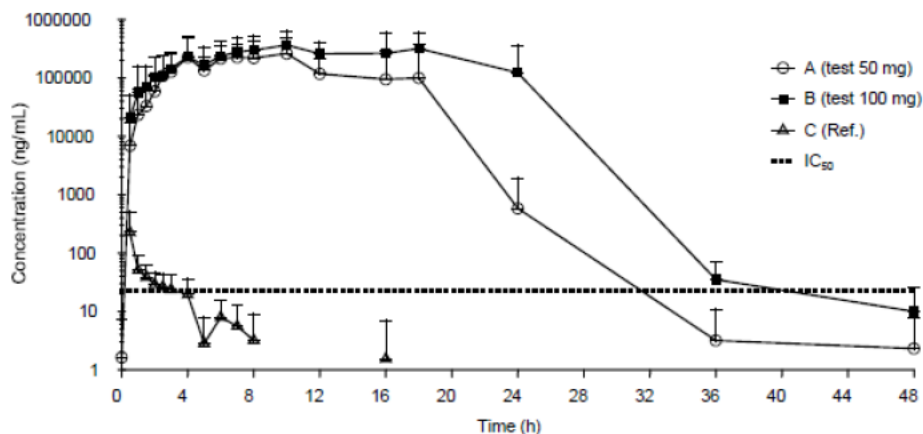
*Frel: relative bioavailability; based on AUC_{0-t} calculated without any dose correction. **PE: point estimate.

: Wilcoxon Signed Rank test

Source: Table 14.4.4-1 and -3/ Appendix 16.2.10.4

The mean saliva concentration-time profiles of the three acyclovir treatments are shown in Figure 3; the individual saliva concentration-time profiles of acyclovir Lauriad® 50 mg (the proposed dose) are shown in Figure 4. After administration of the oral acyclovir tablet, saliva concentrations of acyclovir reached maximal concentrations approximately 0.5 h post-dose, then rapidly declined; salivary concentrations remained above the IC_{50} for three hours. Administration of the Lauriad® formulations resulted in rapid increases in saliva concentrations, reaching peak concentrations 7 and 11 h post-application for the 50 and 100 mg MBT, respectively, and remaining near-constant and high until 18 and 24 h post-application for the 50 and 100 mg MBT, respectively, when concentrations began to decrease. Salivary concentrations remained above the IC_{50} for approximately 32 and 40 h after application of the 50 and 100 mg MBT, respectively. Geometric mean values of salivary C_{max} were 3455- and 4321-fold higher and AUC_{tau} were 12961- and 27374-fold higher for the 50 and 100 mg MBT, respectively, compared to the oral tablet (see Table 2).

Figure 3. Mean $\pm$ SD saliva concentration-time profiles after a single dose of acyclovir 200 mg oral tablet or Lauriad® 50 or 100 mg MBT (log-linear scale; source: Study Report Figure 4)



Source: Table 14.4.2 / Appendix 16.2.10.2

Figure 4. Individual saliva concentration-time profiles after a single dose of acyclovir Lauriad® 50 mg MBT (log-linear scale; source: Study Report Figure 14.2.6.1)

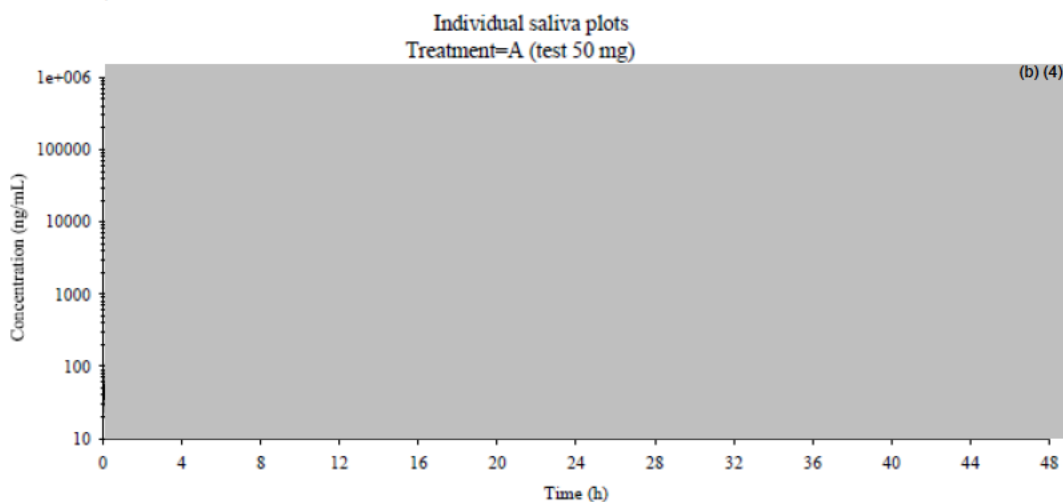


Table 2. Acyclovir salivary pharmacokinetic parameters after a single dose of acyclovir 200 mg oral tablet or Lauriad® 50 or 100 mg MBT (source: Study Report Table 9)

Treatment		t _{lag} (h)	C _{max} (µg/mL)	t _{max} (h)	AUC _{0-t} (h*µg/mL)
Acyclovir	N	12	12	12	12
Lauriad®	Mean	0.00	440	7.95	2900
A: 50 mg	SD	0.00	241	4.08	2400
	Min	0.00	149	3.07	849
	Median	0.00	388	7.04	2090
	Max	0.00	959	18.05	9450
	Geo Mean	NC	387	7.10	2320
	CV%	NC	55	51	83
Acyclovir	N	12	12	12	12
Lauriad®	Mean	0.00	576	11.97	6320
B: 100 mg	SD	0.00	345	4.68	4750
	Min	0.00	158	4.02	2100
	Median	0.00	479	11.08	4900
	Max	0.00	1200	18.03	16700
	Geo Mean	NC	484	11.02	4900
	CV%	NC	60	39	75
Ref.	N	12	12	12	12
Zovirax®	Mean	0.05	0.230	0.73	0.248
C: 200 mg	SD	0.16	0.279	0.50	0.214
	Min	0.00	0.0219	0.52	0.0563
	Median	0.00	0.127	0.53	0.180
	Max	0.57	0.951	2.03	0.736
	Geo Mean	NC	0.112	0.64	0.179
	CV%	346	121	68	86

Source: Table 14.4.4-4 / Appendix 16.2.10.5

The degree of interindividual variability in acyclovir saliva concentrations can be observed in the standard deviations in acyclovir saliva concentrations at each time point (Table 3). For the acyclovir Lauriad® 50 mg MBT (the dose proposed by the Applicant), the mean CV was 150%, with a range of 65 to 347%. These data, as well as the individual acyclovir concentration-time profiles shown in Figure 4 above, confirm that there is extensive variability in acyclovir saliva concentrations following administration of the acyclovir Lauriad® 50 mg MBT.

Table 3. Acyclovir mean $\pm$ SD saliva concentrations after a single dose of acyclovir 200 mg oral tablet or Lauriad® 50 or 100 mg MBT (source: Study Report Table 14.2.2.1)

Time (h)	Treatment A (test 50 mg)			Treatment B (test 100 mg)			Treatment C (Ref.)		
	N	Mean	SD	N	Mean	SD	N	Mean	SD
0	12	1.63	5.66	12	0	0	12	0	0
0.5	12	6880	10500	12	20700	30200	12	227	281
1	12	23400	41000	12	56400	98200	12	50.8	38.3
1.5	12	32900	33300	12	69200	90500	12	38.5	21.9
2	12	58600	53400	12	106000	117000	12	29.3	16.9
2.5	12	108000	131000	12	110000	124000	12	26.3	16.8
3	12	127000	137000	12	144000	125000	12	23.7	17.7
4	12	225000	246000	12	229000	270000	12	19.4	16.9
5	12	134000	87200	12	165000	170000	12	2.82	5.14
6	12	213000	144000	12	236000	177000	12	7.96	7.18
7	12	224000	157000	12	277000	222000	12	5.61	7.12
8	12	218000	196000	12	297000	219000	12	3.19	5.78
10	12	261000	208000	12	364000	269000	12	0	0
12	12	117000	102000	12	259000	145000	12	0	0
16	12	94600	206000	12	262000	326000	12	1.54	5.34
18	12	99500	275000	12	317000	377000	12	0	0
24	12	581	1300	12	122000	237000	12	0	0
36	12	3.18	7.49	12	35.2	33.5	12	0	0
48	12	2.31	5.52	11	9.94	15.0	12	0	0

Both saliva and plasma acyclovir concentrations were less than dose-proportional for C_{max} and more than dose-proportional for AUC_{tau} , although the degree of interindividual variability precludes definitive statements about the dose proportionality of acyclovir Lauriad® tablets from being made.

Acyclovir was not detected in any of the labial samples collected following oral administration of the acyclovir 200 mg tablet. In contrast, acyclovir concentrations were detectable in the earliest labial sample taken after application of the 50 and 100 mg MBT, peaking at approximately 6 h post-application, then decreasing (though remaining detectable) to 12 or 8 h post-application in the 50 and 100 mg MBT treatment groups, respectively. No pharmacokinetic analyses were performed on acyclovir concentrations determined from labial samples.

2.3. DISCUSSION

Study BA2004/21/01 compared the pharmacokinetics of a single dose of 200 mg oral acyclovir tablet or a 50 or 100 mg acyclovir Lauriad® buccal tablet. The Lauriad® formulations provided lower plasma and higher saliva and labial tissue acyclovir concentrations relative to the oral tablet. Although saliva acyclovir concentrations were high shortly after application of the Lauriad® tablet, the degree of intersubject variability was also high across the entire concentration-time profile (see Table 3 and Figure 4). From a clinical pharmacology standpoint, these data support the Applicant's assertion that a BE study (required for a change in manufacturing site) would not be feasible.

3. TRIAL BA2005/21/02 (Pivotal Efficacy Trial Supporting this Application)

A Randomized, Double-Blind, Single Dose, One-Day Early Administration, Multicentre Study Comparing the Efficacy and Safety of Acyclovir Lauriad™ 50 mg Muco-Adhesive Buccal Tablet to Matching Placebo, in the Treatment of Herpes Labialis in Immunocompetent Patients

Coordinating Study Center: Hôpital Henri Mondor (Créteil, France)

Study Dates: 4 Apr to 4 Sep 2009

Report Date: 4 Aug 2011

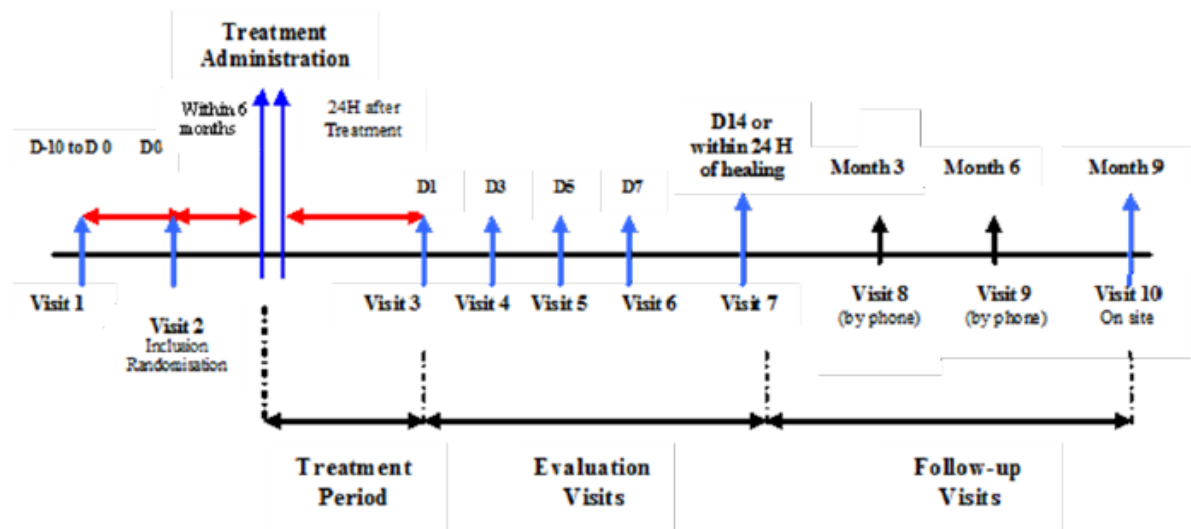
3.1. DESIGN

This trial was a double-blind, randomized, single-dose trial multicenter in 521 subjects with a primary vesicular lesion (MITT population; 378 in the acyclovir Lauriad™ group and 397 patients in the placebo group). Subjects were at least 18 years old and had a history of recurrent labialis lesions, defined as at least 4 episodes of lesions on the cutaneous and/or mucosal surfaces of the lips in the preceding 12 months.

There was one treatment period lasting one day. The treatment consisted of one 50 mg acyclovir Lauriad™ buccal tablet or one matching placebo. The treatment period was patient-initiated within six months of screening, within one hour of the onset of prodromal symptoms and before the appearance of any signs of labial herpes lesions. If the tablet did not adhere for at least six hours, a second tablet was to be applied by the study subject. If no prodromal symptoms were experienced within six months of screening, the subject was excluded from the study.

After application of the buccal tablet, subjects were under evaluation until the healing of the primary lesion or Day 14, whichever came first. Evaluation included a subject diary consisting of a self-questionnaire and a visual analogue scale, which was to be completed by the subject every evening. A clinic visit was also required within 24 h of application of the buccal tablet (Day 1) and on Days 3, 5, 7, and 14 (or within 24 h of healing). Optional follow-up phone calls were conducted 3 and 6 months after healing and an optional visit occurred 9 months after healing. The study schema is shown below.

Figure 1: Study schema (source: Study Report Figure 1)



Saliva sampling for viral titer and acyclovir concentrations were conducted at the visit on Day 1 in selected centers.

The following treatments were not allowed during the study period (application of tablet until healing or Day 14, whichever came first): tablets, suspensions, intravenous formulations, and/or ointments containing antiviral agents known to have activity against HSV-1, HSV-2, HIV, HBV, or HCV; ointments or creams applied to labial or mucosal herpes lesions; NSAIDs or aspirin; steroids or analgesics; topical steroids in the oral area within 4 weeks prior to drug administration; probenecid; or immunosuppressants.

3.2. RESULTS

1721 subjects were randomized to treatment. Of these, 771 received study drug and had complete data on the time and date of study drug application (intent-to-treat [ITT] population) and 521 experienced a primary vesicular lesion and applied study drug (242 in the acyclovir Lauriad™ treatment group and 279 in the placebo group). Of the 521 subjects included in the modified intent-to-treat (mITT) population, 239 patients had aborted primary lesions (130 in the acyclovir Lauriad™ treatment group and 109 in the placebo group).

Thirty subjects did not complete the study, 20 of whom were non-compliant with the protocol. Overall, 68.6% of subjects in the ITT population were female and 94.9% of subjects were Caucasian. The mean age was 41 years, with a range of 18 to 80.

The primary endpoint was time to healing (TTH) of the primary vesicular lesion. The Applicant performed a Log Rank test to assess this endpoint in the mITT population and found that patients in the acyclovir Lauriad™ treatment group experienced a mean shorter duration of TTH of the primary vesicular lesion (7.05 and 7.62 days for the acyclovir and placebo groups, respectively; see Table 4).

Table 4. Time to healing of the primary vesicular lesion after a single dose of acyclovir Lauriad® 50 mg MBT or matching placebo (source: Study Report Table 8)

	ABT 50 mg N=242	Placebo N=279	Total N=521
Patients (N [%])	242 (100.0%)	279 (100.0%)	521 (100.0%)
Events (N)	222	253	475
Censored Observations (N)	20	26	46
Missing Observations (N)	0	0	0
Mean (days) ± SE	7.05 ± 0.18	7.62 ± 0.18	7.36 ± 0.13
Median (days) (95% CI)	7.00 (6.75; 7.31)	7.32 (6.97; 7.92)	7.08 (6.95; 7.40)
Log rank test	0.0150		

Refer to Table 14.2.1.1.1

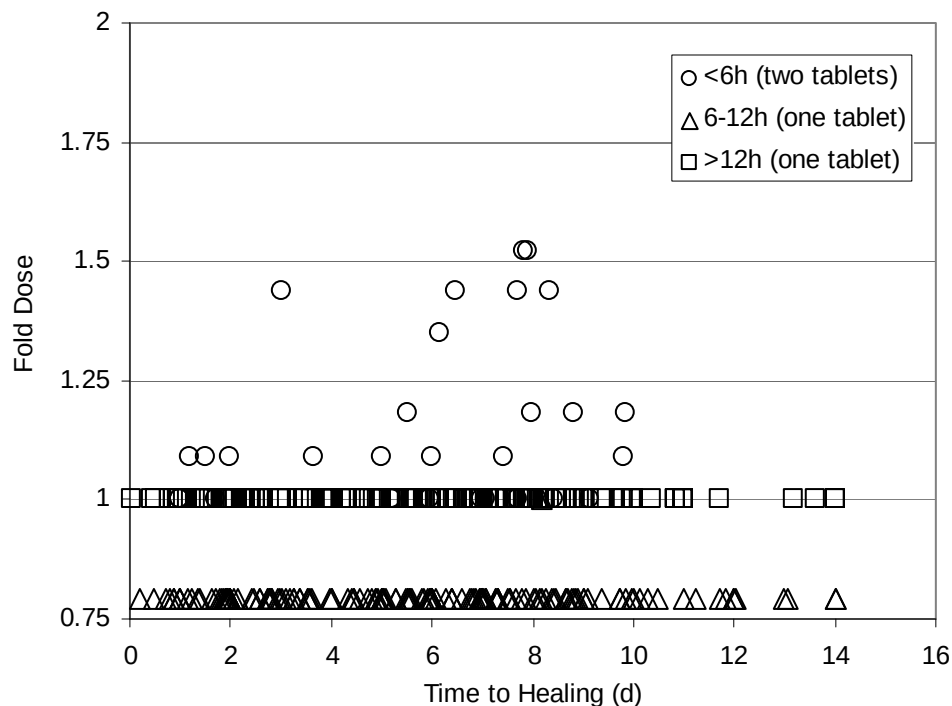
CI: confidence interval; N: number; SE: standard error

Analysis of the same data by the statistical reviewer, Dr. Wen Zeng, determined that the difference in TTH between the acyclovir Lauriad™ and placebo groups was approximately 0.3-0.5 days, depending on the statistical method used. Subgroup analyses examining the effect of duration of tablet adhesion (<6 h, 6-12 h, or >12 h) on TTH revealed that there was no relationship, with the greatest difference in TTH occurring in the >12 h subgroup (0.7 d), followed by the <6 h (0.5 d) and 6-12 h (0.01 d) subgroups, in that order.

A planned secondary endpoint evaluating the relationship between saliva viral titer, saliva acyclovir concentrations, and TTH of the primary vesicular lesion would have provided more detailed information on the exposure-response relationship; however, this endpoint was not evaluated due to limited amount of data collected. A preliminary analysis was performed using in vitro dissolution data and tablet adhesion time to approximate the amount of acyclovir released at the target site (i.e. the buccal cavity).

This analysis relied on the specification detailed in the protocol that the tablet be replaced if it adhered for fewer than 6 h, effectively increasing the possible maximal dose administered to about 75 mg acyclovir (given approximately 50% tablet dissolution after 6 h, irrespective of pH). The exact duration of adhesion of the first tablet was recorded if it was less than 6 h; otherwise, it was recorded as 6-12 h or >12 h. The adhesion time of the second tablet was not recorded. Therefore, in the <6 h subgroup, the dissolution of the first tablet was approximated based on in vitro data, and it was assumed that the second tablet dissolved completely to give one-fold of the applied dose. For the 6-12 h and >12 h subgroups (both of which applied a single tablet), 0.79- and 1-fold dose estimates were used, based on in vitro dissolution data. The relationship between the estimated fold dose and TTH were plotted (see Figure 2). The lack of a clear correlation suggests that the estimated dose delivered (and thereby the amount of acyclovir available at the site of action) is not associated with TTH.

Figure 2: Relationship between estimated fold dose of acyclovir and the time-to-healing of the primary vesicular lesion



3.3. DISCUSSION

Study BA2005/21/02 evaluated the effect of a single dose of acyclovir Lauriad™ 50 mg on the time-to-healing of the primary vesicular lesion in immunocompetent patients with a history of recurrent herpes labialis lesions. Although local concentrations were not quantified, in theory, tablet adhesion time may be used as a surrogate for the amount of acyclovir delivered locally and subjects with longer adhesion times should have shorter TTH compared to those with shorter adhesion times. However, no relationship between adhesion time and TTH was observed, suggesting that local acyclovir exposures were either not sufficient to achieve significant antiviral activity or that the applicant's postulated mechanism of drug reaching the site of the cold sore (licking of the lips) is not well-supported.

Although a clear relationship between the estimated amount of acyclovir in the buccal cavity and TTH was not identified with the data submitted, this lack of a relationship does not preclude the approval of this product from a clinical pharmacology perspective. A statistically significant and robust difference in TTH between the acyclovir Lauriad™ and placebo groups was not found by the statistical reviewer and for this reason, this study does not support approval.

4. LABELING RECOMMENDATIONS

Not applicable.

5. CONCLUSIONS

The Applicant proposed a transfer in product manufacturing from the current site (at which clinical batches were made) to a new site (at which commercial batches will be produced). Per the Agency's SUPAC-MR guidance, this is a Level 3 change requiring chemistry and dissolution documentation as well as a single-dose bioequivalence study in plasma. The pharmacokinetic data reviewed in this memorandum were submitted by the Applicant to support its claim that saliva acyclovir concentrations (which are more meaningful to the efficacy of this product than are plasma acyclovir concentrations) are highly variable within and between subjects due to their dependence on saliva flow. These data (including the individual and mean concentrations, as well as the standard deviations, of saliva acyclovir concentrations measured in the 12 subjects for whom pharmacokinetic data were available from Study BA2004/21/01) were reviewed in this memorandum and appear to support the Applicant's request for a waiver of the BE study required for a change in manufacturing site.

The relationship between estimated acyclovir dose delivered to the buccal cavity and time-to-healing of the primary vesicular lesion in Trial BA2005/21/02 was evaluated. There was no clear evidence of a significant relationship between tablet adhesion time and time-to-healing, based on the limited amount of data available for analysis. However, a clearly demonstrated dose-response (or exposure-response) relationship is not needed for approval of this product from a clinical pharmacology perspective; provided robust efficacy is demonstrated in one or more clinical trials. In future efficacy trial(s), the sponsor may want to explore higher doses or a different dosing frequency in order to increase the likelihood of demonstrating more robust efficacy.

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/s/

LESLIE W CHINN
12/07/2012

SHIRLEY K SEO
12/07/2012

Office of Clinical Pharmacology New Drug Application Filing and Review Form

General information about the submission	
NDA/BLA Number	203791 (SDN 1/SN 0000)
OCP Division	DCP4
Medical Division	Antivirals
OCP Reviewer	Leslie Chinn, Ph.D.
OCP Team Leader	Shirley Seo, Ph.D. (acting)
Pharmacometrics Reviewer	To be determined
Date of Submission	12 Mar 2012
OCP Review Estimated Due Date	
Medical Division Due Date	7 Dec 2012
PDUFA Due Date	11 Jan 2013

General information about the drug/biologic	
Brand Name	Sitavig
Generic Name	Acyclovir
Drug Class	Nucleoside analogue against herpes simplex virus
Indication(s)	Local treatment of orofacial herpes and (b) (4)
Dosage Form	50 mg mucoadhesive buccal tablet (Lauriad®)
Dosing Regimen	50 mg single dose
Route of administration	Buccal
Sponsor	BioAlliance Pharma
Priority Classification	Standard

Clinical pharmacology and biopharmaceutics information				
Study Type	Incl. at Filing	No. of Studies Submitted	No. of Studies Reviewed	Critical Comments
Table of Contents incl. reports, tables, data	☐			
Tabular Listing incl. all human studies	☒			
Human PK Summary	☒			
Labeling	☒			
Reference Bioanalytical and Analytical Methods	☒			Plasma, saliva, tissue strips

CLINICAL PHARMACOLOGY AND BIOPHARMACEUTICS
Filing Form/Checklist for NDA/BLA or Supplement

I. CLINICAL PHARMACOLOGY				
Mass Balance	☐			
Isoenzyme Characterization	☐			
Blood/Plasma Ratio	☐			
Plasma Protein Binding	☐			
Pharmacokinetics (e.g. Phase 1)				
<i>Healthy Volunteers</i>				
Single Dose	☒	1		Conducted to compare bioavailability of the 50-mg and 100-mg acyclovir buccal tablet to the acyclovir 200-mg oral tablet. The results from this study were also used to support a request for a waiver of BE study requirements (based on the SUPAC-MR guidance on change in manufacturing site) due to high interindividual variability in PK and thus, infeasibility of conducting a meaningful BE study.
Multiple Dose	☐			
<i>Patients</i>				
Single Dose	☐			
Multiple Dose	☐			
<i>Dose Proportionality – Fasting/Non-Fasting</i>				
Single Dose	☐			
Multiple Dose	☐			
<i>Drug-Drug Interaction Studies</i>				
In Vivo Effects on Primary Drug	☐			
In Vivo Effects of Primary Drug	☐			
In Vitro	☐			
<i>Special Populations</i>				
Ethnicity	☐			
Gender	☐			
Pediatrics	☐			
Geriatrics	☐			
Renal Impairment	☐			

CLINICAL PHARMACOLOGY AND BIOPHARMACEUTICS
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Hepatic Impairment	☐			
<i>Pharmacodynamics</i>				
Phase 2	☐			
Phase 3	☐			
<i>Pharmacokinetics/Pharmacodynamics</i>				
Proof of Concept (Phase 1 or 2)	☐			
Clinical Trial (Phase 3)	☐			
<i>Population Analyses</i>				
Data-rich	☐			
Data-sparse	☐			
II. BIOPHARMACEUTICS				
<i>Bioavailability</i>				
Absolute Bioavailability	☐			
Relative Bioavailability (solution as reference)	☐			
Relative Bioavailability (alt. formulation as ref.)	☒	1		See Pharmacokinetics section above for study description.
<i>Bioequivalence</i>				
Traditional Design (single/multiple dose)	☐			
Replicate Design (single/multiple dose)	☐			
<i>Food-Drug Interaction</i>	☐			
<i>Biowaiver Request (based on BCS class)</i>	☐			
<i>Dissolution (alcohol-induced dose-dumping)</i>	☐			
III. OTHER CLINICAL PHARMACOLOGY/BIOPHARMACEUTICS				
<i>Genotype/Phenotype</i>	☐			
<i>Chronopharmacokinetics</i>	☐			
<i>Pediatric Development Plan</i>	☐			
<i>Literature References</i>	☐			
TOTAL NUMBER OF STUDIES		1		

On initial review of the NDA/BLA application for filing:				
Content Parameter	Yes	No	N/A	Comment

CLINICAL PHARMACOLOGY AND BIOPHARMACEUTICS
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Criteria for Refusal to File (RTF)					
1	Has the applicant submitted bioequivalence data comparing to-be-marketed product(s) and those used in the pivotal clinical trials?	☐	☐	☐	
2	Has the applicant provided metabolism and drug-drug interaction information?	☐	☐	☒	
3	Has the sponsor submitted bioavailability data satisfying the CFR requirements?	☒	☐	☐	
4	Did the sponsor submit data to allow the evaluation of the validity of the analytical assay?	☒	☐	☐	
5	Has a rationale for dose selection been submitted?	☒	☐	☐	
6	Is the clinical pharmacology and biopharmaceutics section of the NDA organized, indexed and paginated in a manner to allow substantive review to begin?	☒	☐	☐	
7	Is the clinical pharmacology and biopharmaceutics section of the NDA legible so that a substantive review can begin?	☒	☐	☐	
8	Is the electronic submission searchable, does it have appropriate hyperlinks and do the hyperlinks work?	☒	☐	☐	
Criteria for Assessing Quality of an NDA (Preliminary Assessment of Quality)					
Data					
9	Are the data sets, as requested during pre-submission discussions, submitted in the appropriate format (e.g., CDISC)?	☒	☐	☐	
10	If applicable, are the pharmacogenomic data sets submitted in the appropriate format?	☐	☐	☒	
Studies and Analyses					
11	Is the appropriate pharmacokinetic information submitted?	☒	☐	☐	
12	Has the applicant made an appropriate attempt to determine reasonable dose individualization strategies for this product (i.e., appropriately designed and analyzed dose-ranging or pivotal studies)?	☐	☒	☐	
13	Are the appropriate exposure-response (for desired and undesired effects) analyses conducted and submitted as described in the Exposure-Response guidance?	☐	☐	☒	
14	Is there an adequate attempt by the applicant to use exposure-response relationships in order to assess the need for dose adjustments for intrinsic/extrinsic factors that might affect the pharmacokinetics or pharmacodynamics?	☐	☐	☒	
15	Are the pediatric exclusivity studies adequately designed to demonstrate effectiveness, if the drug is indeed effective?	☐	☐	☒	

CLINICAL PHARMACOLOGY AND BIOPHARMACEUTICS
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16	Did the applicant submit all the pediatric exclusivity data, as described in the WR?	☐	☐	☒	
17	Is there adequate information on the pharmacokinetics and exposure-response in the clinical pharmacology section of the label?	☐	☐	☒	
General					
18	Are the clinical pharmacology and biopharmaceutics studies of appropriate design and breadth of investigation to meet basic requirements for approvability of this product?	☒	☐	☐	
19	Was the translation (of study reports or other study information) from another language needed and provided in this submission?	☐	☐	☒	

IS THE CLINICAL PHARMACOLOGY SECTION OF THE APPLICATION FILEABLE?

☒ **Yes** ☐ **No**

If the NDA/BLA is not fileable from the clinical pharmacology perspective, state the reasons and provide comments to be sent to the Applicant.

Please identify and list any potential review issues to be forwarded to the Applicant for the 74-day letter.

Leslie Chinn, Ph.D.	23 Apr 2012
Reviewing Clinical Pharmacologist	Date
Shirley Seo, Ph.D.	23 Apr 2012
Acting Team Leader/Supervisor	Date

This is a representation of an electronic record that was signed electronically and this page is the manifestation of the electronic signature.

/s/

LESLIE W CHINN
06/01/2012

SHIRLEY K SEO
06/04/2012